SSD Class Activity

Terminal Mastery CTF

10 Challenges · 110 Points · Pure Bash

NO EXTENSIONS. NO EXCEPTIONS.

- **Deadline: Oct 11, 11:59 PM. Hard cutoff.**
- No extensions. No "my internet was slow." No "I didn't see the message."
- No partial marks. Either the flag is correct or it isn't.
- After Oct 11 11:59 PM IST, submissions will not be counted.
- **This is the only announcement. There will be no reminders.**

What is this?

A Capture The Flag (CTF) activity where you solve 10 terminal challenges using bash commands. Each challenge gives you a `FLAG{...}` that you submit for points.

Every challenge is different per roll number. Copying someone else's commands or flags will **NOT** work. Your flags are generated from your roll number.

How to Set Up (do this FIRST)

Just ONE command. That's it.

```
bash ssd_class_activity.sh
```

It will ask for your **roll number**. Enter it. That's it.
Your challenges are now set up at `~/ctf_arena/`.
The server URL is already built into the script — you don't need to set anything.

WARNING

- The script **deletes itself** after running. You cannot re-read it.
- If you need to re-run it, re-download the file.

How to Solve Challenges

There are 10 levels. You can attempt them in **any order**.

```
cd ~/ctf_arena/level_01          # go to a level
cat README.txt                    # read YOUR challenge
# ... solve it using bash commands ...
# ... you'll find a FLAG{...} ...
```

How to Submit a Flag

THIS IS THE MOST IMPORTANT PART. READ CAREFULLY.

When you find a flag (it looks like `FLAG{abc123def456}`), submit it like this:

```
~/ctf_arena/submit 1 FLAG{abc123def456}
```

Breaking it down:

`~/ctf_arena/submit`	The submit command. You MUST use the full path.
`1`	The level number (1 through 10).
`FLAG{abc123def456}`	The flag you found. Copy-paste it exactly.

WARNING

Common mistakes that will waste your time:

- `submit 1 FLAG{...}` ← **WRONG.** You need the full path.
- `~/ctf_arena/submit 1 FLAG{...}` ← **CORRECT.**
- `./submit 1 FLAG{...}` ← Also works if you're inside `~/ctf_arena/`
- Do NOT add quotes around the flag.
- Do NOT add spaces inside `FLAG{...}`.
- The flag is case-sensitive. Copy-paste, don't type.

If the flag is correct:

```
Level 1 solved!  (+5 pts)
Score: 1 / 10  |  Time: 3m
```

If the flag is wrong:

```
Wrong flag for Level 1.
```

How to Check Your Score

Run this anytime to see which levels you've solved and your total points:

```
~/ctf_arena/score
```

It shows a scorecard like this:

```
01.  Hidden in Plain Sight +5 pts
02.  Needle in a Haystack 5 pts
03.  Deep Underground +10 pts
...
Total: 15 / 110 pts
```

Levels and Points

Level	Name	Points	Skill Tested
01	Hidden in Plain Sight	5	Finding hidden files
02	Needle in a Haystack	5	Searching text in files
03	Deep Underground	10	Finding files in directories
04	Column Extraction	10	Filtering rows and extracting columns from CSV
05	The Substitution	10	Find-and-replace in text
06	Permission Denied	10	File permissions
07	The Pipeline	15	Chaining multiple commands together
08	Line Hunter	10	Finding specific lines and fields
09	Decode the Layers	15	Encoding/decoding (base64, reverse)
10	The Grand Finale	20	All of the above combined
	Total	**110**	

You can attempt levels in **any order**. Start with the easy ones (1, 2) if you're unsure.

WARNING

Do this yourself. Don't use AI.

- Do not paste challenge files into ChatGPT, Claude, Gemini, Copilot, or any AI tool.
- The challenges have decoy files that **will give AI the wrong answers**. If you rely on AI, you'll submit wrong flags and waste your time.
- This activity tests YOUR terminal skills. The whole point is for you to learn the commands.
- **What's allowed:** Using `man <command>`, Googling command syntax, reading documentation.
- **What's not:** Having any AI solve it for you.

Rules

1. **No partial marks.** A flag is either correct or wrong. There is no "close enough."
2. **No extensions.** Deadline is 11:59 PM tonight. After that, nothing counts.
3. **Challenges are unique per roll number.** The file names, search keywords, column numbers, and encoding layers are all different for each student. Copying commands from someone else will give you the wrong answer.
4. **Flags are unique per roll number.** Even if you somehow get someone else's flag, it won't work for your roll number.
5. **The READMEs tell you WHAT to find, not HOW.** You need to figure out which terminal command to use. That's the whole point.
6. **No hints command.** Figure it out yourself or use `man <command>`.
7. **Time is tracked.** Your solve times are logged.

Works On

- **Linux** (Ubuntu, etc.) — works out of the box
- **WSL** (Windows Subsystem for Linux) — works out of the box
- **Mac** (macOS Terminal) — works out of the box
- **Windows CMD/PowerShell** — does **NOT** work. Use WSL.

Quick Reference (the full flow)

```
# 1. Run the setup
bash ssd_class_activity.sh

# 2. Go to a level
cd ~/ctf_arena/level_01

# 3. Read YOUR challenge
cat README.txt

# 4. Solve it (use terminal commands!)
# ... you'll find FLAG{something} ...
```

```
# 5. Submit
~/ctf_arena/submit 1 FLAG{something}

# 6. Check your score
~/ctf_arena/score

# 7. Move to next level
cd ~/ctf_arena/level_02
cat README.txt
# ... repeat ...
```

NO EXTENSIONS. NO EXCEPTIONS.

Deadline: Oct 11, 11:59 PM.

No extensions. No exceptions. No excuses.

Start now.